Foundation model embeddings from intratumoral and peritumoral regions on T2-weighted MRI for classification of testicular germ cell tumors

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1. DESCRIPTION OF PURPOSE

Testicular germ cell tumors are rare cancers and classified as seminoma germ cell tumors (SGCTs) and nonseminoma germ cell tumors (NSGCTs). This distinction is clinically important because SGCT and NSGCT have different prognostic and treatment pathways [1,2]. MRI provides high tissue contrast for testicular lesion characterization and can depict tumor morphology, internal signal features, and adjacent parenchyma [3]. However, qualitative MRI assessment may be limited when SGCT and NSGCT show overlapping appearances.

Radiomics converts medical images into quantitative descriptors of shape, intensity, and texture [4,5]. Standardized radiomic feature definitions have been advanced through the Image Biomarker Standardisation Initiative and their extraction through software toolkits, such as CaPTk and PyRadiomics [6,7,9]. In oncologic imaging, analysis of both the tumor and surrounding peritumoral tissue may capture complementary information related to local tissue response, remodeling, compression, edema, and tumor-microenvironment interaction [8]. Prior testicular MRI radiomics studies have investigated SGCT vs NSGCT classification, however, they have not extensively analyzed the peritumoral region of focal testicular lesions on imaging [10-12]. Besides, for rare cancers with small datasets, systematic evaluation of multiple AI approaches including handcrafted radiomics, convolutional neural networks (CNN) and pre-trained MRI foundation models have not been explored.

The purpose of this study was to evaluate whether intratumoral and peritumoral T2-weighted MRI biomarkers could support noninvasive SGCT vs NSGCT classification. The primary analysis used frozen MRI-CORE ViT-B foundation model [13] embeddings as learned MRI representations from intratumoral and peritumoral regions. Interpretable radiomics and ResNet50 embeddings were included as comparison baselines. The study was organized into intratumoral, peritumoral, and intratumor-peritumor fusion experiments. The novelty of the work is the matched comparison of foundation model embeddings, radiomics, and CNN embeddings across the same MRI regions and patient level cross validation splits.

2. METHODS

2.1 Dataset, Tumor masks and peritumoral ring generation

This retrospective secondary analysis used a publicly available *2021_FelicianiGiacomo_Collection1* dataset which contains testicular T2-weighted MRI studies with histopathologic ground truth for SGCT and NSGCT[11]. N= 42 patient studies and 44 focal testicular lesions were available in this dataset. After strict SGCT/NSGCT endpoint filtering, T2-weighted imaging and mask availability checks, feature processing requirements, peritumoral shell quality control, and matched intratumor-peritumor requirements, the final analysis cohort included 23 TGCT patients, consisting of 16 SGCT and 7 NSGCT cases. The binary endpoint was SGCT vs NSGCT, coded as SGCT=0 and NSGCT=1.

Tumor regions of interest were defined on T2-weighted MRI using expert annotations on the dataset [10,11]. Testicular parenchyma masks were used to generate distance-based peritumoral shells around the tumor boundary. Three concentric peritumoral rings were generated (0-3 mm, 3-5 mm, and 5-7 mm). Each shell was restricted to the testicular parenchyma mask and explicitly constrained to exclude tumor voxels, reducing intratumoral leakage into the peritumoral analysis.

2.2 Foundation-model and CNN embedding analysis

For analysis, frozen pretrained encoders were used in feature extraction mode. MRI-CORE was evaluated as an MRI specific foundation model encoder [13], while ResNet50 served as an ImageNet-pretrained CNN baseline [15,16]. Tumor and peritumoral T2-weighted MRI crops were resized to 224 x 224 pixels. MRI-CORE was loaded through a ViT-B-

compatible architecture; mean pooling was used for the intratumoral branch and max pooling for the peritumoral branch, producing 768-dimensional patient level embeddings.

Foundation model and ResNet50 embeddings were reduced using PCA fit within each training fold, 5 principal components were retained for each intratumoral and peritumoral embedding branch and were then used as an input to class balanced logistic regression to generate held out NSGCT probabilities. MRI-CORE embeddings were 768 dimensional and ResNet50 embeddings were 2048 dimensional before PCA.Intratumor-peritumor fusion branches used held out probability and did not involve any additional PCA or logistic regression fitting. Intratumoral, peritumoral, and intratumor-peritumor fusion branches were evaluated for both MRI-CORE (F_{int}, F_{peri}, F_{comb}) and ResNet50 (R_{int}, R_{peri}, R_{comb}).

2.3 Radiomics feature extraction and modeling

Intratumoral radiomic features (C_{int}) and peritumoral radiomic features (C_{peri}) were extracted from expert tumor masks and peritumoral rings, respectively. Intratumoral candidate features include first-order intensity, shape, GLCM/Haralick, GLRLM, GLSZM, GLDM, NGTDM, and filtered radiomics features. The intratumoral analysis evaluated 1,794 candidate features. Peritumoral radiomics included 88 base features per shell, producing 264 peritumoral features across the three rings.

Feature selection and model fitting were performed within training folds. Candidate radiomic features were screened using Mann-Whitney U testing, effect size filtering, and correlation filtering. Classification used median imputation, standard scaling, and L1-regularized logistic regression with class-balanced weighting. Model performance was evaluated using repeated stratified 3-fold x 20-repeat cross validation splits, and out-of-fold predicted NSGCT probabilities were used for AUC, sensitivity, specificity, fusion, and probability separation analysis. All models used the same patient level repeated stratified 3-fold x 20-repeat split assignments to support paired comparison across radiomics, MRI-CORE, and ResNet50.

2.4 Probability level fusion and statistical analysis

Fusion was performed at the held-out probability level. The final best region fusion combined the intratumoral model with the peritumoral model. For patient *i* in cross-validation split *s*, Let $C_{1,i,s}$ represent the intratumoral predicted NSGCT probability and $C_{2,i,s}$ represent the peritumoral predicted NSGCT probability, the late-fused probabilities were calculated as follows;

$$C_{\mathrm{LF},i,s} = \frac{C_{1,i,s}+C_{2,i,s}}{2}$$

where $C_{LF,i,s}$ is the late-fused predicted NSGCT probability. This formulation was applied separately to MRI-CORE, radiomics and ResNet50 outputs using held out predictions only.

Equal-weight probability averaging was used instead of feature-level or learned fusion to limit overfitting in the small cohort, preserve regional interpretability, and avoid feature-selection/model-fitting stage. Each fused probability combined predictions for the same held-out patient and split, so neither component model had been trained on that patient.

Model performance was summarized using AUC with standard deviation and two-sided Mann-Whitney U testing of out-of-fold predicted NSGCT probabilities. Sensitivity, specificity, and balanced mean were calculated at a fixed probability threshold of 0.5 for direct cross-model comparison. Because only 7 NSGCT cases were included, secondary sensitivity-oriented operating points were reported separately. Derived fusion probabilities were not entered as independent multivariable covariates.

2.5 Exploratory segmentation analysis

Expert tumor masks were used as reference masks for primary analysis. An exploratory nnU-Net segmentation experiment was also performed using manually annotated T2-weighted MRI and ADC images to segment tumor regions [14]. In this small cohort, nnU-Net achieved a Dice score of approximately 0.90. This exploratory result suggests that future automated workflows could use nnU-Net-derived tumor masks for tumor detection, followed by distance-transform-based generation of peritumoral rings.

3. RESULTS

The final matched cohort included 23 patients (16SGCT, 7NSGCT).All MRI-CORE(F_{int},F_{peri},F_{comb}) radiomics(C_{int},C_{peri},C_{comb})and ResNet50(R_{int},R_{peri},R_{comb}) models were evaluated using the same 60 patient level cross-

validation splits, with each patient appearing in a held-out fold 20 times (Table 1). C_{int} showed significant intratumoral discrimination, and C_{comb} achieved the highest radiomics AUC through equal-weight intratumor-peritumor probability fusion. C_{peri} showed high NSGCT sensitivity and low significance. F_{int} achieved the strongest overall single-region performance, with AUC 0.911 ± 0.062 and significant SGCT-NSGCT probability separation (P=.001). F_{comb} remained significant and improved SGCT specificity. ResNet50 embeddings were evaluated as a conventional deep-learning baseline. R_{int} showed the best single region CNN performance, while R_{comb} remained significant and preserved high SGCT specificity.

Table 1. Quantitative performance evaluation across all assessed approaches.

**P values are from within model Mann-Whitney U tests comparing patient level OOF predicted NSGCT probabilities between true SGCT and true NSGCT cases and are not pairwise AUC comparison between respective models.*

Model	Region / strategy	AUC Mean ± SD	Sensitivity NSGCT	Specificity SGCT	Balanced accuracy	P value*
C_{int}	**Intratumoral radiomics**	**0.821 ± 0.116**	**0.857**	**0.750**	**0.804**	**.015**
C_{peri}	Peritumoral radiomics	0.696 ± 0.114	1.000	0.625	0.813	.154
C_{comb}	**Equal weight intratumor-peritumor fusion**	**0.848 ± 0.087**	**0.857**	**0.625**	**0.741**	**.008**
F_{int}	**MRI-CORE intratumor, mean pooling**	**0.911 ± 0.062**	**0.857**	**0.750**	**0.804**	**.001**
F_{peri}	MRI-CORE peritumor, max pooling	0.679 ± 0.116	0.571	0.750	0.661	.198
F_{comb}	**Equal weight MRI-CORE fusion**	**0.875 ± 0.077**	**0.714**	**0.813**	**0.763**	**.004**
R_{int}	**Intratumoral Resnet50 embedding**	**0.812 ± 0.112**	**0.571**	**0.813**	**0.692**	**.018**
R_{peri}	Peritumoral Resnet50 embedding	0.679 ± 0.116	0.571	0.688	0.629	.198
R_{comb}	**Equal weight intratumor peritumor Resnet50 fusion**	**0.786 ± 0.109**	**0.571**	**0.813**	**0.692**	**.033**

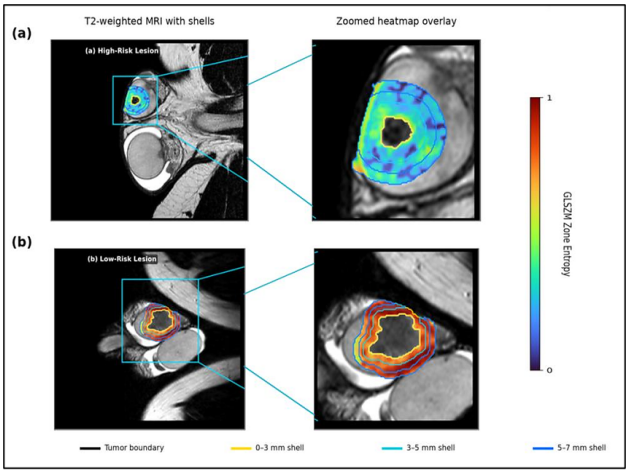


Figure 1. Peritumoral radiomic heatmap visualization. Representative T2-weighted MRI examples show tumor boundaries, magnified tumor-adjacent regions, and concentric peritumoral rings. The 0-3 mm, 3-5 mm, and 5-7 mm rings were restricted to testicular parenchyma and constrained to exclude tumor voxels. Heatmap overlays illustrate spatial variation in peritumoral texture using a GLSZM Zone Entropy-based radiomic feature.

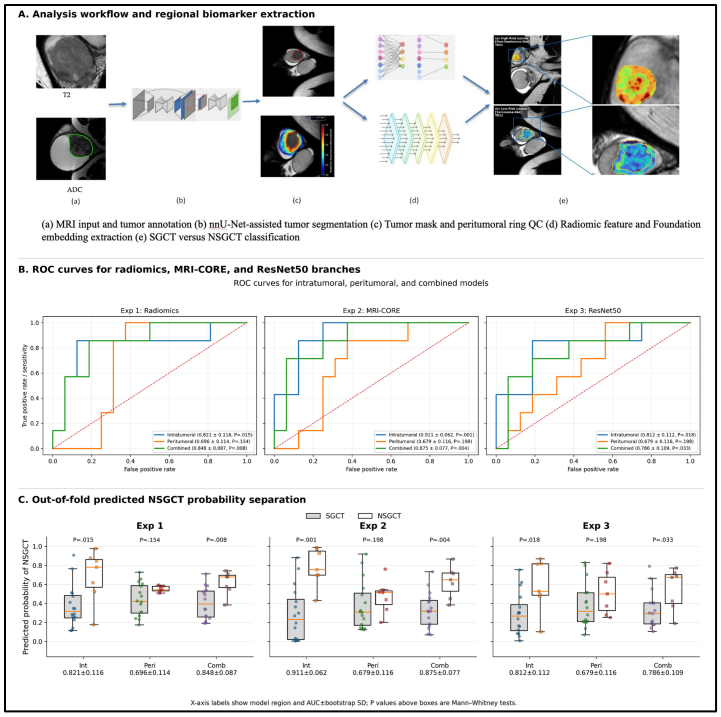


Figure 2. Integrated workflow and model-evaluation summary. (A) MRI input, tumor annotation, segmentation-assisted ROI preparation, feature/embedding extraction, and SGCT versus NSGCT classification workflow. (B) Receiver operating characteristic curves for radiomics, MRI-CORE foundation-model embeddings, and ResNet50 embeddings across intratumoral, peritumoral, and combined branches. (C) Patient-level out-of-fold predicted NSGCT probability distributions for SGCT vs NSGCT cases; P values above boxes correspond to two-sided Mann-Whitney U tests.

5. CONCLUSIONS

This study evaluated intratumoral and peritumoral T2-weighted MRI biomarkers for noninvasive SGCT vs NSGCT classification in a final matched cohort of 23 patients. The analysis compared three complementary representations: interpretable radiomics, frozen MRI-CORE ViT-B foundation-model embeddings, and ResNet50 CNN embeddings. Radiomics used handcrafted first-order, shape, and texture descriptors, while MRI-CORE provided latent learned embeddings that may collectively encode morphology, intensity, texture, and contextual information. All models were evaluated using identical repeated stratified 3-fold x 20-repeat patient-level cross-validation splits.

Radiomics tumor-peritumor fusion achieved the best radiomics performance (C_{comb}: AUC 0.848 ± 0.087; P=.008), supporting complementary value from peritumoral information. MRI-CORE intratumoral embeddings showed the strongest performance (F_{int}: AUC 0.911 ± 0.062; P=.001), while MRI-CORE fusion remained significant (F_{comb}: AUC 0.875 ± 0.077; P=.004) and improved SGCT specificity. ResNet50 served as a conventional baseline and performed below MRI-CORE and radiomics fusion in the combined comparison. These findings suggest that MRI-CORE captures a strong learned intratumoral signal and that tumor-peritumor fusion can provide complementary information for noninvasive TGCT subtype classification. Larger independent cohorts are required for external validation.

6. BREAKTHROUGH WORK

The breakthrough aspect of this study is the matched, region aware comparison of an MRI-CORE, Radiomics and Resnet50 baseline models, across the same intratumoral and peritumoral T2-weighted MRI regions. Instead of using only intratumoral features, this study tests whether the tumor adjacent microenvironment represented by parenchyma restricted and peritumoral rings adds complementary information to the intratumoral biomarkers. This provides preliminary evidence for noninvasive TGCT subtype characterization that combines learned embeddings with biologically interpretable regional imaging context.

DISCLOSURE OF PRIOR OR CONCURRENT SUBMISSIONS

No prior or concurrent submission: this work is original work that has not been previously published and is not currently under review elsewhere.

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